

Chris Zhang

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EDUCATION

Northeastern University, College of Engineering

Master of Science Degree in Mechanical Engineering, Mechatronics Concentration

Courses: Advanced Mechanics of Materials, Materials Process Selection, Math Methods, Mechatronic Systems, Control Systems, Scientific Machine Learning, Power Electronics
Extracurriculars: ASME, Club Ultimate Frisbee Captain

Boston, MA

April 2026

GPA: 3.63

Bachelor of Science Degree in Mechanical Engineering

Awards: Dean's List (Fall 2021, Spring 2022, Spring 2024, Spring 2025), Dean's Scholarship

Relevant Courses: Heat Transfer, Fluid Mechanics, Thermodynamics, Systems and Controls, Measurements, Electrical Engineering, Capstone

Extracurriculars: Society of Asian Scientists and Engineers, Electric Racing, ASME

May 2025

GPA: 3.5

EXPERIENCE

ICO Energy and Engineering, Boston, MA

Commissioning Engineer

July 2024 – December 2024

- Performed site inspections and executed commissioning checklists for HVAC systems across 6+ commercial facilities, verifying safety compliance and system operation according to specifications
- Collected and tabulated pipe flow and energy data on site for meter allocation services, highlighting faulty meters that needed calibration to correct building energy use data to reduce excess energy spending by ~15%
- Prepared technical report in Excel modeling building energy usage intensity from energy and water billing to gauge potential ~20% savings from commissioning work for potential client
- Tracked air source heat pump performance across outdoor temperature ranges and compared efficiency against conventional heating and cooling systems to determine optimal heat pump operation times

Instron, Norwood, MA

Applications Engineer

July 2023 – December 2023

- Determined optimal tensile test equipment based on knowledge of ISO and ASTM standards for specific client applications regarding plastics, metals, thin films, elastomers, and more
- Generated reports sent to high profile clients detailing recommended testing frames, fixtures, and extensometers, assisting with orders up to \$40,000+
- Engaged in multi-disciplinary collaboration to generate novel testing report template for 3D printed plastic samples to be employed for potential clients seeking 3D printed applications

PROJECTS

Northeastern University, Boston, MA

January 2026 – April 2026

Canine Rehabilitation Wheelchair, Wearable Robotics

- Collaborated with group of 4 to create an assistive mobility device for hind-leg paralyzed canines that generated passive range of motion during wheelchair locomotion
- Designed and fabricated 3D-printed pedal assemblies in SolidWorks, iterating designs based on patient-specific measurements and constraints from crankshaft rotation
- Processed IMU trial data in MATLAB using high-pass filtering and kinematic analysis to quantify hip, knee, and back articulation, validating benefits of pedal-driven hind leg motion

Northeastern University, Boston, MA

June 2024 – April 2025

Auto Return Trainer, Mechanical Engineering Capstone

- Co-led team of 6 to design and fabricate an automated soccer ball launcher delivering game-realistic shots at speeds up to 70 mph for goalkeeper training
- Designed motor mounts and wheel axle adapters in SolidWorks with ± 0.005 " tolerances and GD&T specifications, ensuring precision alignment for consistent ball launch trajectories
- Managed \$3000 project BOM in Excel and coordinated testing facility reservations and measuring equipment availability with Capstone advisors

SKILLS

CAD & Design: SolidWorks (CSWA Certified), AutoCAD, Abaqus, GD&T, DFM

Computer & Electrical: Excel, PowerPoint, Word, Arduino, Digital Multimeter, Oscilloscope, LTspice, PLECS

Fabrication & Prototyping: 3D Printing, Bench Assembly, Hand Tools, Soldering, Sensor Integration

Programming Languages: MATLAB, Python C++, Java, JavaScript, C#